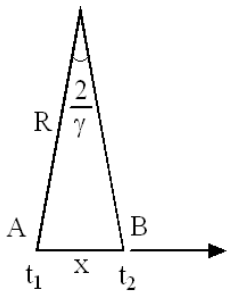
3：

P

$2/\gamma$



$$x=\frac{2R}{\gamma}. \qquad (1)$$

$\beta=v/c$

$$t_2-t_1=\frac{2R}{\gamma c\beta}. \qquad (2)$$

A B

$$t'=\frac{2R}{\gamma c}, \qquad (3)$$

$$\Delta T=t_2-t_1-t'. \qquad (4)$$

$$\gamma=\frac{mc^2}{m_0c^2}=\frac{1}{\sqrt{1-\beta^2}}, \qquad (5)$$

$1-\beta\to 0$

$$\frac{2}{\beta}(1-\beta)=\frac{1}{\gamma^2}. \qquad (6)$$

(4)

$$\Delta T=\frac{R}{c\gamma^3}. \qquad (7)$$

$$\Delta T=9.51\times10^{-20}\text{s} \qquad (8)$$